Chapter 4

Artifact: value-density of bounded strategies

Bounded strategies are value-dense under (A1) and (A6)

On a general probability space, near-optimal strategies need not have integrable hedge ratios, so Hornik's theorem cannot be applied directly. This is failure point (F1). We repair it under two hypotheses from Chapter 3: (A1) requires that the liability Z is essentially bounded, i.e. $Z \in L^\infty$, and (A6) requires that every admissible gain is nonnegative, i.e. $W(\delta) \geq 0$ a.s. for all $\delta \in \mathcal{H}$. Together they give $X + W(\eta) = -Z + W(\eta) \geq -\|Z\|_\infty$ for every admissible η, which is the deterministic lower bound needed for Lemma 4.4. The next lemma shows that, under (A1) and (A6), bounded strategies are value-dense: every strategy can be replaced by a bounded one without worsening the OCE objective by more than an arbitrary $\varepsilon > 0$. In particular, π equals the infimum of the objective over bounded strategies alone.

Lemma 4.1. *Assume (A1) and (A6), and let $X = -Z$. Then for every $\delta \in \mathcal{H}^{\mathrm{u}}$ and every $\varepsilon > 0$ there exists a bounded $\delta' \in \mathcal{H}^{\mathrm{u}}$ with*

$$\rho_\ell\big(X + W(H \circ \delta')\big) \leq \rho_\ell\big(X + W(H \circ \delta)\big) + \varepsilon.$$

In particular,

$$\pi(X) = \inf\big\{\rho_\ell\big(X + W(H \circ \delta)\big) : \delta \in \mathcal{H}^{\mathrm{u}},\ \delta \textit{ bounded}\big\}.$$

Proof. Fix $\delta \in \mathcal{H}^{\mathrm{u}}$ and write $\eta := H \circ \delta$, $Y := X + W(\eta)$. Define the componentwise truncations $\delta^{(j)} = (\delta_k^{(j)})_{k=0,\dots,n-1}$ for $j \in \mathbb{N}$ by

$$\delta_k^{(j)} := \begin{cases} \delta_k & \text{if } \delta_k \in [-j, j], \\ -j & \text{if } \delta_k < -j, \\ j & \text{if } \delta_k > j, \end{cases} \qquad k = 0, \dots, n-1$$

(applied componentwise if $d > 1$). Then each $\delta^{(j)}$ is adapted (hence in $\mathcal{H}^{\mathrm{u}}$) and bounded by j. Set $\eta^{(j)} := H \circ \delta^{(j)}$ and $Y_j := X + W(\eta^{(j)})$.

Since there are finitely many time steps, there exists a (random) index $j_0 < \infty$, namely

$$j_0 := \max_{\substack{0 \le k \le n-1 \\ 1 \le i \le d}} \lceil |\delta_k^i| \rceil,$$

such that $\delta_k^{(j)} = \delta_k$ for all k and all $j \ge j_0$. In particular $\delta_k^{(j)} \to \delta_k$ pointwise for every k. The recursive definition of $H \circ$ then yields $\eta^{(j)} \to \eta$ pointwise, and therefore $Y_j \to Y$ pointwise.

By (A6), $W(\eta^{(j)}) \ge 0$ a.s., so $Y_j \ge -\|Z\|_\infty$ a.s. by (A1). Lemma 4.4 with $B = \|Z\|_\infty$ gives

$$\limsup_{j \to \infty} \rho_\ell(Y_j) \le \rho_\ell(Y).$$

Choose j large enough that $\rho_\ell(Y_j) \le \rho_\ell(Y) + \varepsilon$, and set $\delta' := \delta^{(j)}$. $\square$